1. Idea Name

BizMitra — An AI-powered business decision support system for small and medium-sized enterprises (SMEs) to forecast sales, manage inventory effectively, and receive real-time stock alerts.

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2. Authors' Names

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3. Idea Description / Problem Statement

Challenge Faced:
Many small and medium-sized enterprises (SMEs) struggle to maintain an accurate balance between inventory and sales.
Due to limited technical expertise and lack of affordable tools, they often overstock (leading to wastage and increased holding costs) or understock (resulting in lost sales and poor customer satisfaction).
A critical part of this problem is the lack of real-time visibility into stock levels, leading to delayed responses to potential stockouts or overstock situations.

Use Case Description:
A unified platform that provides accurate sales forecasts, inventory recommendations, and real-time stock alerts, enabling SMEs to make proactive, data-driven decisions.

Target Audience:
Small and medium business owners, retail stores, wholesalers, and local suppliers who lack access to expensive ERP or analytics tools.

How It Solves the Problem:
BizMitra leverages AI-driven forecasting, a clear visual dashboard, and a proactive Stock Alert System to:

- Predict future sales trends
- Suggest optimal stock levels
- Provide real-time notifications for low stock, overstock, and stockout scenarios
- Prevent stockouts and overstocking
- Improve revenue planning and reduce operational risks

4. Proposed Solution / Idea

Big Picture Vision:
Build a centralized, user-friendly platform that gives real-time business insights, demand forecasting, actionable inventory management recommendations, and critical stock-level notifications.

POC-Specific Scope & Approach:

- Use existing sales history to train a SARIMAX time-series model for forecasting
- Use business rules and thresholds to build an inventory management engine
- Implement a configurable Stock Alert System to monitor inventory against thresholds and trigger notifications
- Integrate the model outputs with an interactive dashboard
- Provide an AI-powered chatbot (Gemini) to answer user queries in natural language

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5. Technology Stack

A. Frontend Development (User Interface)

- **React**: For building the interactive user interface
- **Vite**: For fast builds and development server
- **Tailwind CSS**: For modern, responsive UI styling
- **Recharts**: For displaying sales forecasts and KPIs
- **PapaParse**: To parse CSV forecast data from Supabase

B. Backend & Data Platform

- **Supabase Postgres Database**: To store and query all historical sales data and user-configured alert thresholds
- **Supabase Storage**: To store forecast CSVs and trained ML models
- **Supabase Client (supabase-js):** To communicate with the database from frontend
- **Supabase Realtime**: To listen for database changes and push instant stock alerts to the UI

C. Artificial Intelligence & Machine Learning

- **Python**: For offline training of forecasting models
- **Pandas**: For cleaning and preparing data

- **Statsmodels (SARIMAX):** For time-series forecasting
- **Scikit-learn**: For feature engineering
- **Joblib**: For saving and loading trained models
- **Google Gemini API**: For powering the AI chatbot's natural language understanding

D. Deployment & DevOps

- **Git & GitHub:** For version control and code hosting
- **Vercel:** For deploying the live React application

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6. Target Customers

- Local SMEs and shop owners
- Retailers and wholesalers
- Distributors managing multiple products and regions
- Small businesses without existing ERP systems

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7. Key Benefits to Target Customers

- **Accurate Forecasting:** Predict sales trends to plan ahead
- **Inventory Optimization:** Avoid stockouts and overstocking
- **Proactive Monitoring:** Receive instant alerts for critical inventory changes, enabling immediate action
- **Reduced Costs:** Minimize holding and wastage costs
- **Better Cash Flow Management:** Align purchases with demand
- **Data-Driven Decision Making:** Make smarter business decisions quickly
- **User-Friendly:** Easy to use without technical knowledge

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8. Affordability

1. **Use of Open-Source Technologies**
   BizMitra leverages open-source frameworks like **Python, React, Tailwind, and Statsmodels**, which eliminates costly software licensing fees. This makes the platform

accessible to SMEs without burdening them with recurring tool costs.

2. **Cloud-Native and Lightweight Infrastructure**
   By using **Supabase (Postgres + Storage)** and deployment services like **Vercel**, BizMitra avoids the need for heavy on-premise infrastructure. SMEs can access insights with minimal setup and pay only for what they use.

3. **Focused Feature Set with Maximum Impact**
   Instead of bloated features found in large ERP systems, BizMitra focuses on **sales forecasting, inventory recommendations, and stock alerts** — the core needs of SMEs. This ensures **maximum value at minimal operational cost**.

4. **Low Learning Curve and No Specialized Staff Required**
   The intuitive dashboard, natural-language chatbot, and easy visualization tools remove the need for hiring data analysts or IT experts. SMEs can use the system directly, saving significant costs in training and manpower.

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9. Technical Consideration / Functional Flow Architecture

A. Data & Model Preparation

Data Collection & Preprocessing Approach

- **Sourcing Reliable Data from Kaggle**
  We started with an inventory sales dataset from Kaggle, ensuring a broad mix of attributes like **store, product, region, units sold, price, discount, demand, and seasonality** to capture realistic retail dynamics.

- **Handling Missing & Inconsistent Records**
  Missing values, duplicate records, and inconsistencies (like mismatched units or product IDs) were carefully treated using **imputation techniques, standardization, and data validation checks** to ensure reliability.

- **Feature Engineering for Business Insights**
  Derived meaningful features such as **Revenue, Profit Margin, Days of Supply, Demand Fulfillment Ratio, Lag Features (7/28 days), and Rolling Statistics**. These transformations captured both **short-term demand spikes** and **long-term seasonal patterns**.

- **Capturing Temporal & Seasonal Trends**
  Introduced **time-based features** (Week of Year, Month, Quarter, Is Month-End) to model seasonality effectively, ensuring the forecasting model adapts to **holiday peaks,**

off-season dips, and quarterly cycles.

- **Incorporating External Influences**
  Enriched the dataset with **Weather Conditions, Promotion Indicators, and Epidemic Events** to make the model more robust and explainable in real-world unpredictable situations.

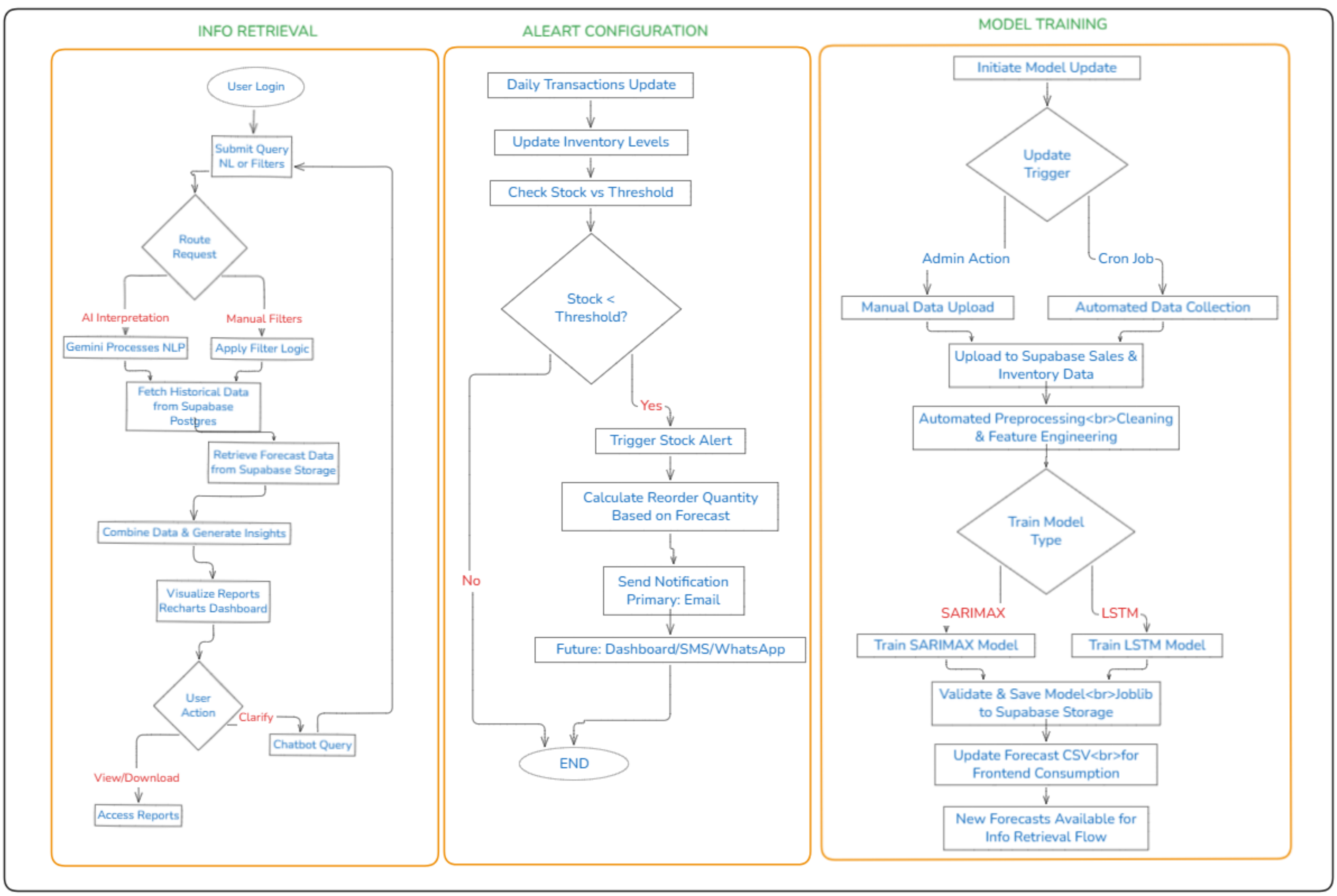
- **Data Normalization & Outlier Treatment**
  Applied **scaling and outlier detection techniques** to stabilize skewed sales distributions ensuring smoother model convergence and improved accuracy.

- **Model Training:**
  - Use Statsmodels (SARIMAX) and Scikit-learn to train a time-series model on the cleaned data
- **Model Saving:**
  - Save the trained model as a `.pkl` file using Joblib for reuse
- **Forecast Generation:**
  - Load the saved model and generate a 90-day future forecast of sales and demand
  - Save this forecast as a `.csv` file
- **Threshold Configuration:**
  - Store user-defined thresholds (e.g., "alert me when product X stock < 20") in the Supabase database
- **Storage & Versioning:**
  - Push the code to GitHub
  - Upload the forecast `.csv` file and trained model file to Supabase Storage

B. Functional Flow Architecture (Updated with Realtime Alerts)

C. Live Application User Flow

- **Frontend Delivery:**
  - The React application is served to the user's browser via Vercel
- **Initial Data Fetch:**
  - Fetch complete historical dataset from Supabase Postgres
  - Fetch forecast `.csv` from Supabase Storage
  - Fetch user's configured alert thresholds
- **Visualizations Rendered:**
  - Recharts renders KPIs and charts based on fetched data



- **Filtering by User:**
  - User applies filters (date range, region, product line, etc.)

  - React sends filtered queries to Supabase Postgres
  - UI re-renders based on the new dataset
- **Real-Time Alert Triggered:**
  - An inventory update occurs (e.g., a sale is made)
  - Supabase Realtime pushes this update to the frontend
  - The app checks the new stock level against thresholds and instantly displays a visual and audible alert if needed
- **AI Chatbot Query:**
  - User asks a question (e.g., "What is the forecast for electronics next month?")
  - Question is sent to Google Gemini API
  - Gemini identifies if the query is about:
    - Historical data → Fetch from Supabase Postgres
    - Forecasted data → Filter loaded forecast `.csv`
    - Current stock alerts → Check the active alerts state in the UI
  - Sends result to Gemini for summarization
  - Chatbot displays answer in plain English
- **Forecast Download:**
  - User clicks "Download"
  - Browser downloads `forecast_output.csv` directly from Supabase Storage
- **Logout:**
  - User logs out
  - Supabase securely ends the session and terminates the realtime subscription

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10. Observations

The performance of BizMitra is evaluated using key parameters:

- Forecast accuracy is measured using MAPE and RMSE
- SARIMAX provides explainable and reliable results for short-term predictions
- Stock alert notifications are tested for real-time responsiveness and low latency
- Consistent accuracy in predicting demand patterns ensures reliability in decision-making

11. Future Scope

1. **Shift from SARIMAX to LSTM for Smarter Forecasting**

   - Upgrade the forecasting engine from traditional SARIMAX models to **deep learning–based LSTM models**, enabling more **accurate, dynamic, and adaptive predictions** that improve as more data flows in.

2. **Intelligent & Automated Inventory Control**
   - Extend the stock alert feature into a **fully automated notification and recommendation engine**, delivering **real-time alerts via Email, SMS, or WhatsApp**, and even **auto-generating purchase orders** when critical thresholds are breached.
3. **Product-wise & Multi-Store Forecasting**
   - Enhance forecasting granularity by enabling **category-level and product-level demand forecasting**, and extend support for **multiple stores and SKUs**, making BizMitra scalable for growing businesses.
4. **AI-Driven Reinforcement Learning Policies**
   - Introduce **reinforcement learning models** to dynamically **optimize inventory policies**, ensuring that reorder levels, buffer stock, and alert thresholds continuously adapt to changing market and demand patterns.
5. **Global Expansion with Multi-Language & Marketing Optimization**
   - Empower businesses across regions by enabling **multi-language support**, and extend capabilities to include **marketing spend optimization**, linking sales forecasts with promotional campaigns for maximum ROI.

6**. Analytics Tool Expansion**

- **Custom Dashboards & Advanced Visualization**: Allow users to design their own dashboards, drill down into cohort/funnel analysis, and view predictive trends similar to tools like Power BI.
- **Cross-Platform Integrations**: Support seamless import/export with systems like **Excel, Shopify, WooCommerce, and QuickBooks**, making BizMitra a plug-and-play analytics hub.
- **AI-Driven Prescriptive Insights & Mobility**: Go beyond queries by suggesting proactive actions (e.g., *"Increase stock of electronics before festival season"*) and deliver mobile-first dashboards for real-time alerts on-the-go.

12. Challenges Faced

1. **Data Consistency & Cleaning**
   - Collecting and cleaning sales data from multiple sources to ensure **accuracy, consistency, and reliability** was a major challenge, as noisy or missing records

could directly affect forecasting quality.

2. **Building an Explainable Yet Accurate Model**
   - Striking the right balance between **accuracy and interpretability** was crucial. We needed a forecasting model that not only produced strong predictions but also remained **transparent and explainable** to business users.
3. **Handling Seasonal Demand & Volatility**
   - Managing **seasonal spikes, sudden demand fluctuations, and unpredictable events** posed a big challenge, requiring careful feature engineering and adaptive model design.
4. **Designing Reliable Real-Time Alerts**
   - Building an alert system that was **timely and actionable**, without generating excessive or noisy notifications, was difficult yet essential to support inventory decision-making.
5. **Balancing Cost with System Performance**
   - Ensuring the platform remained **cost-efficient, lightweight, and fast** while handling real-time queries and visualizations was a constant engineering challenge.

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13. Conclusion

BizMitra offers a practical, low-cost, and scalable AI solution for small and medium businesses by using lightweight time-series forecasting (SARIMAX) with explainable inventory logic, an intuitive dashboard, and a proactive real-time Stock Alert System to generate clear restock recommendations. This integrated approach helps reduce stockouts and excess inventory, improve cash flow, and enable owners to take immediate action. Built with React, Supabase, and Python, its modular design ensures easy upgrades and quick deployment, while its focus on affordability, data privacy, explainability, real-time responsiveness, and user-friendly design makes it ideal for non-technical shop owners.
